

Homework — 1.3.4 Web Technologies

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.3.4 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Part of the HTML for a web page is shown below.

```
<h1>Court Booking</h1>
<p>Book courts up to seven days ahead.</p>
<p>Members may bring one guest.</p>
<div id="warning">The sports hall is closed this week.</div>
```

(a) The developer wants **every** paragraph shown in blue text. State the type of CSS selector that should be used. **[1]** (AO1)

(b) The single warning box is to be styled differently from everything else. State whether a `.class` selector or an `#id` selector is the more appropriate choice for it, and justify your answer. **[2]** (AO2)

2. Explain why the styling for a large website is placed in an external **CSS** file rather than being written directly into the HTML of every page. **[2]** (AO2)

3. Describe the role of a **web crawler** in how a search engine builds its index. **[2]** (AO1)

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4. The **PageRank** algorithm ranks pages by importance.

(a) Explain how an incoming link from a **high-ranking** page affects a page's rank differently from a link from a **low-ranking** page. [2] (AO1)

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(b) State what is meant by saying PageRank is calculated **iteratively** until it **converges**. [2] (AO1)

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5. A web form lets a user choose a username. JavaScript in the browser checks instantly that the box is not empty (**client-side**), and the server then checks that the username is not already taken in the database (**server-side**).

(a) Give **one** benefit of doing the "not empty" check **client-side**. [1] (AO2)

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(b) Explain why the "username already taken" check **must** be done server-side. [2] (AO2)

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Part B — Retrieval: keep it fresh (~6 marks)

6. (1.3.3) State the difference between a **switch** and a **router**, referring to the type of address each uses. [2] (AO1)

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7. (1.3.1) Explain **one** reason why **symmetric** encryption has a key distribution problem that **asymmetric** encryption avoids. [2] (AO1)

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8. (1.3.2) A relational database contains a **Customer** table and an **Order** table. State the purpose of a **primary key**, and state how a **foreign key** links the two tables. [2] (AO1)

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